EVA KATO

evakato14@gmail.com

EDUCATION

Utrecht University 2023 – 2025
MSc. in Computer Science – Game and Media Technology Programme (cum laude) Utrecht, NL
- Relevant courses: Advanced Computer Graphics, Geometric Algorithms, Optimization/Vectorization, Computer Vision

Purdue University 2020 – 2023
BSc. in Computer Science West Lafayette, IN
- Relevant courses: Numerical Methods, Analysis of Algorithms, Systems Programming, Computer Graphics, Machine Learning

PUBLICATIONS

Eva Kato, Ningna Wang, Silvia Sellán, and Shane Scott. SDF-Based Separation Constraints for Feature-Mapping Topology Optimization. *ACM Symposium on Computational Fabrication (SCF), Poster Session, 2026*.

Eva Kato, Tim Ophelders, Alex Telea, and Jiří Kosinka. Simplification of Locally Refined Gradient Meshes. *Graphical Models, 2026*.

RESEARCH EXPERIENCE

Braid Technologies Sep '25 – Present
Lead Research Engineer, Geometry Processing Tokyo, JP
- Lead a core geometry R&D team building tooling that turns design intent directly into manufacturable engineering parts
- Own the technical direction of geometry kernel and the team's research roadmap, unifying multiple geometric representations
- Built a high-performance C++ geometry engine for in-loop design optimization, exposed to internal DSLs for research workflows
- Contributed to differentiable, physics-informed topology and shape optimization methods for structural applications
- Partner across teams to ship production-grade software integrated with simulation pipelines on HPC systems
- Built working prototypes of automated design optimization with leading Japanese automotive and industrial manufacturers

Columbia University Sep '25 – Nov '25
Visiting Researcher – Advisor: Prof. Silvia Sellán New York, NY
- Enforced manufacturability in topology optimization by restricting the design space to a library of parametrized primitives
- Formulated geometric constraints as differentiable JAX energies for joint discrete–continuous optimization over primitives

MIT Summer Geometry Initiative Jul '25 – Aug '25
Research Fellow – Mentors: Bailey Miller, Justin Solomon, Anh Truong Remote, USA
- Developed harmonic control variates for variance reduction in Monte Carlo geometry processing (Walk-on-Spheres)
- Improved numerical stability of the log-heat method, reducing error from floating-point underflow at long distances
- Formulated convex shape optimization problems in terms of support functions, enabling efficient optimization over convex bodies

Braid Technologies Apr '25 – Jun '25
Research Intern, Geometry Processing – Mentor: Shane Scott Tokyo, JP
- Engineered and validated geodesic path solvers that generate stiffening-rib layouts on engineering parts
- Implemented robust intrinsic triangulations for improving accuracy of surface PDE solvers and computing geodesics

Utrecht University Jun '24 – Feb '25
Graduate Student – Advisors: Prof. Tim Ophelders, Jiří Kosinka, Alex Telea Utrecht, NL
- Implemented state-of-the-art vector graphics representation supporting T-junctions and its renderer, which allows local refinement with fewer control points than standard gradient meshes (grids of bicubic Bézier patches)
- Developed simplification methods leveraging this representation, reducing patch count while preserving visual appearance

Purdue University May '23 – Aug '23
Undergraduate Research Assistant – Advisor: Voicu Popescu West Lafayette, IN
- Developed view-dependent level-of-detail methods to reduce rendering load for VR clients

AWARDS

MIT Fellowship, Summer Geometry Initiative 2025
Honorable Mention, High-Performance Graphics (HPG) Student Competition 2025
Selected Work, Electronic and Time-Based Art, Purdue Art & Design Juried Undergraduate Exhibition 2023

VOLUNTEER EXPERIENCE

MIT Summer Geometry Initiative, Volunteer — 2026
World Cube Association, Competition Organizer — 2017, 2018
World Cube Association, Competition Volunteer — 2014 – Present

OTHER EXPERIENCE

Purdue University, School of Languages and Culture — Mar '23 to Jul '23
Software Developer (Contract) — West Lafayette, IN

- Built and deployed a VR application for Japanese language acquisition research, used in multiple user studies

Warner Bros. Discovery — Jun '22 to Aug '22
Software Developer Intern — Atlanta, GA

NVC Logistics — Jun '19 to Aug '19
Software Developer Intern — Rockleigh, NJ

SKILLS

Programming: C++, Python (JAX, PyTorch, OpenCV), C#, Javascript/Typescript, LaTeX, GLSL/HLSL
Frameworks/Libraries: Vulkan, OpenGL, Three.js, CMake, Git
Applications: Unity, ParaView, Visual Studio, NVIDIA Nsight
Languages: English (native), Japanese (professional working)